- The ICD might not have full RAM retention when it is in an idle state.
- When the ICD is powered off to change the battery.
- Power or network outage causing the connection between the client and the ICD to be interrupted.
- The client is unavailable for any reason (e.g. during a software update or hosted on a mobile device that is sometimes out-of-home).

The Check-In message is sessionless and relies on a shared secret that has been given to the ICD during the registration of the client using the ICD Management cluster.

Use Case Requirements

The ICD Check-In Protocol representation of the Check-In Counter SHALL be the ICD Counter. It SHALL follow all the requirements of the Check-In Counter.

The ICD Check-In Protocol representation of the Check-In Protocol persistent data store SHALL be the RegisteredClients attribute in the ICD Management cluster. The attribute SHALL store all the client registration information.

The Application Data for the ICD Check-In Protocol use case SHALL be the Active Mode Threshold attribute in the ICD Management cluster. Before being used in the encryption process, the Active Mode Threshold SHALL be converted into a 2-byte little-endian value.

Protocol State Machine

The state machine diagram represents the states an ICD can be in. The diagram represents the state of the ICD for one client or MonitoredSubject. The ICD can be in a different state for each client simultaneously.

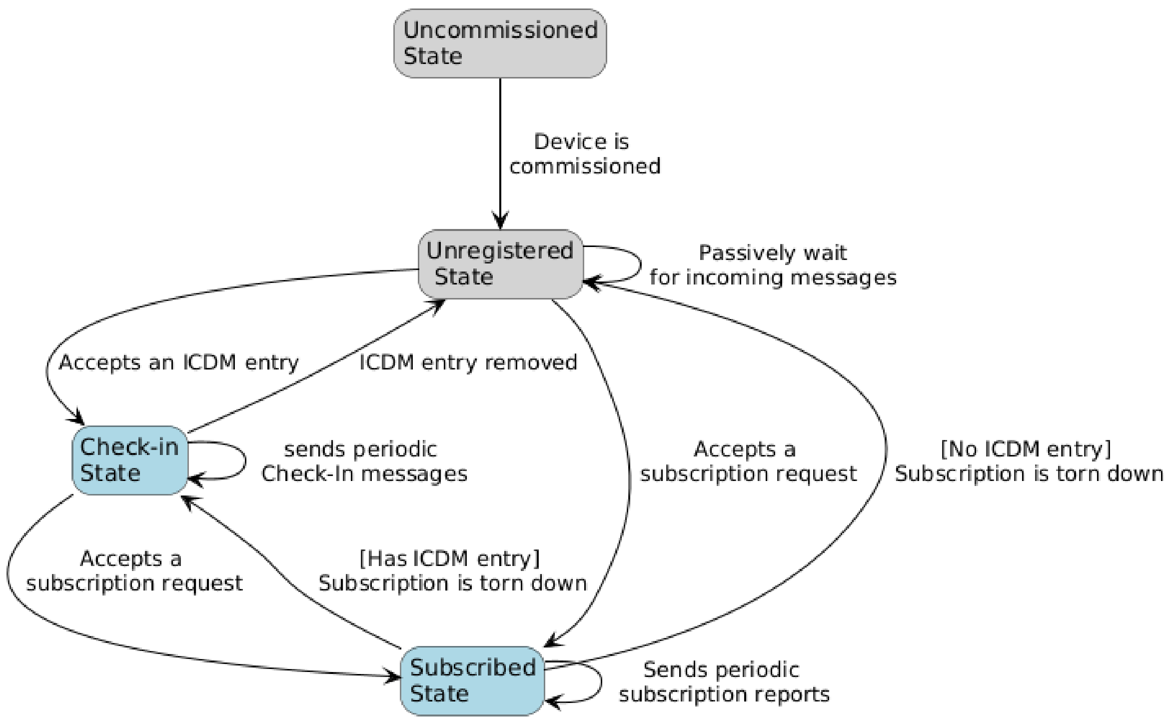

Figure 77. ICD State Machine